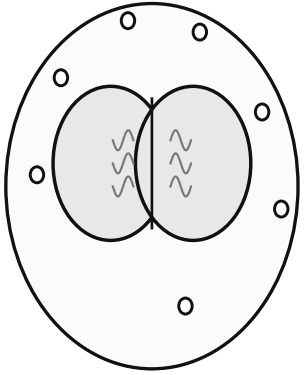
From neural activity to EEG evidence

Measurements constrain the hypothesis; they do not directly reveal a new physical field.

A | Neural generators



Scalp electrodes record potential differences

B | Forward measurement

Neural current distributions + tissue conductivity

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Reference-dependent EEG channels + noise + measurement artifacts

Inverse reconstruction is non-unique

C | Reported observables

- Evoked potential amplitude and latency
- Oscillatory power and phase structure
- Coupling statistics and model scores

Conceptual schematic. Neural sources, extracranial fields, and derived EEG statistics must not be treated as equivalent entities.